

Econ 2 - Lecture 16 - 5/27/26

Lecture Quiz 8 (Short) Released Today, Due Monday, 6/1

Friday Review Session in NH 2212, this week only!

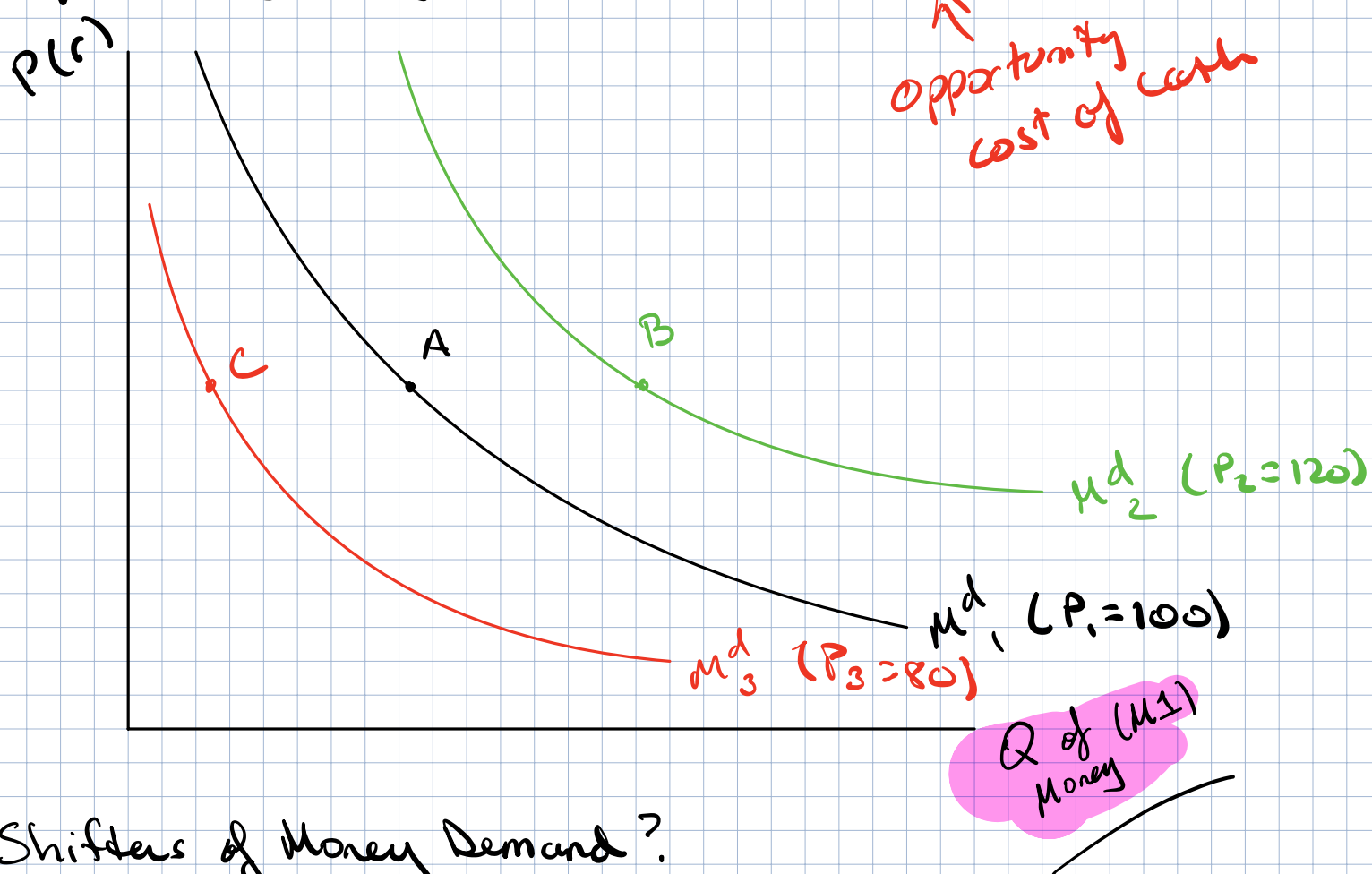
Discussion Activity #5 Next Week (Week 10)

Final Exam  $\Rightarrow$  Tuesday, June 9<sup>th</sup> at 12<sup>pm</sup>

40 Questions  $\rightarrow$  Expectations  $\rightarrow$  Rising Complexity into Week 10

Last Class: Money Demand

Money Demand Curve ( $M^d$ ): Relationship between quantity of money demanded and interest rate (price of money)



Shifters of Money Demand?

1. Prices = CPI = GDP Deflator \*

Price Level  $\uparrow \rightarrow$  Households need more cash  $\uparrow M^d$

2. Size of Wealth Box

# Money Supply: Who determines the money supply?

## Federal Reserve

→ Open market operations

→ Set  $M^s = 1 \text{ Trillion}$  (Cash in circulation  $\approx 1 \text{ Tril}$ )

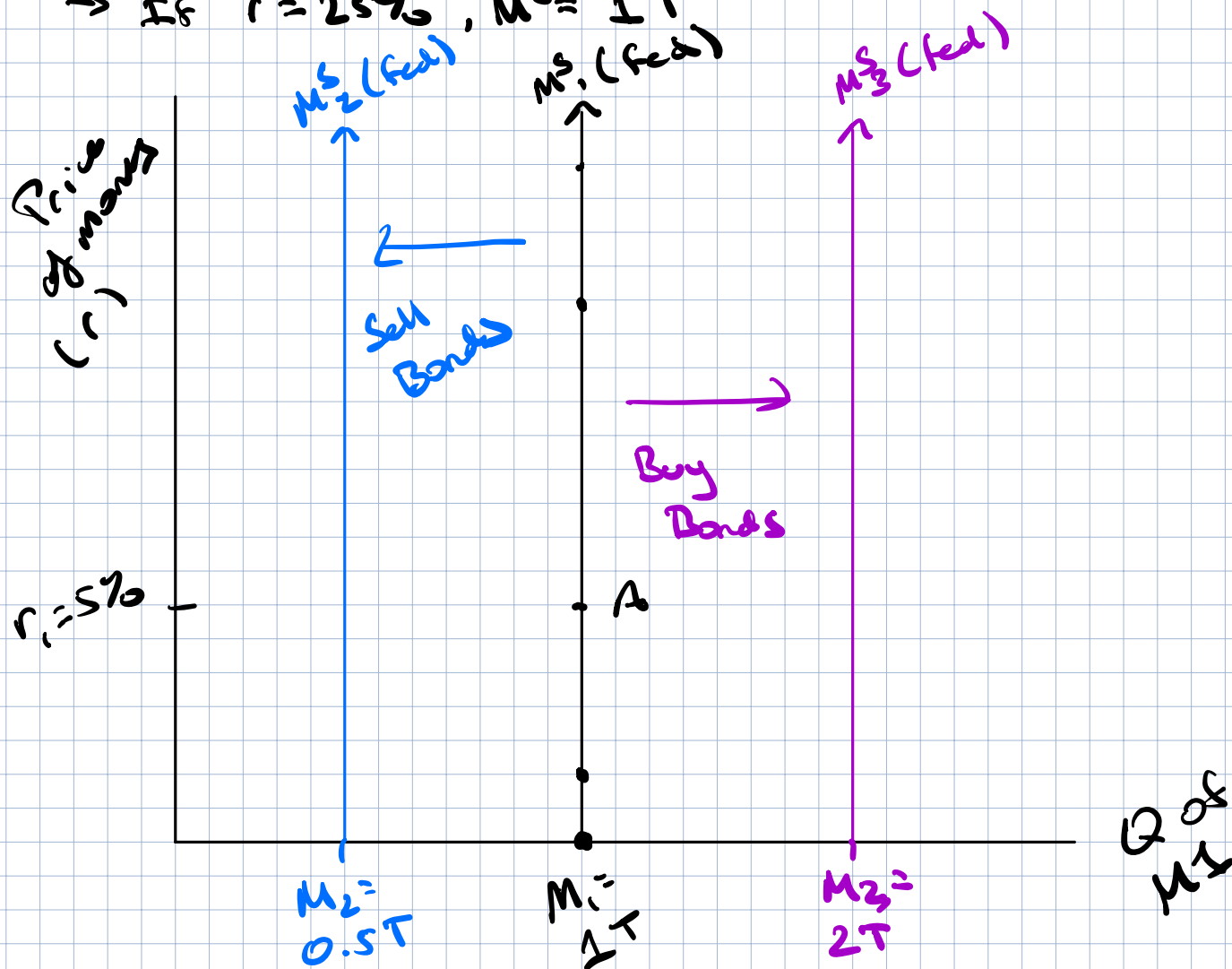
→ Quantity

→ If  $r = 0\%$ ,  $M^s = 1T$

→ If  $r = 5\%$ ,  $M^s = 1T$

→ If  $r = 25\%$ ,  $M^s = 1T$

Perfectly inelastic supply curve



Fed wants to decrease  $M^s \rightarrow$  Action?

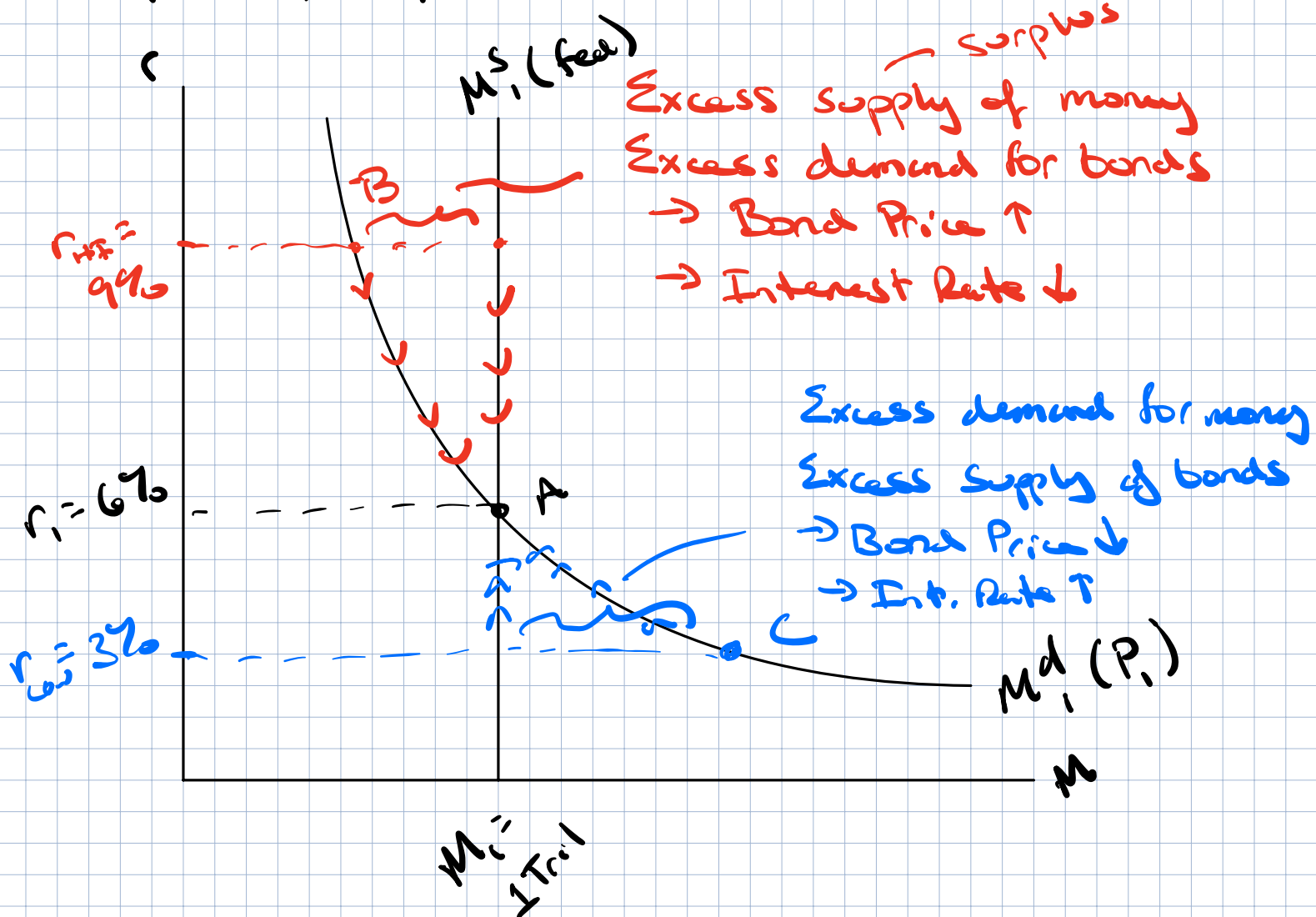
Buy or Sell Bonds to Public

Sell bonds to public  $\rightarrow$  Public gives up cash, for bonds

Increase  $M^s \rightarrow$  Buy bonds from public

Combine  $M^d \equiv M^s$

$$M^d_1(P_1), M^s_1(\text{fed}) = 1 \text{ Tril}, r_1 = 6\%$$



Equilibrium at Pt. A  $\Rightarrow$

Fed set  $M^s = 1 \text{ Tril}$ , the Quantity of  $M_1$  demanded is 1 Tril at  $r_1 = 6\%$

Imagine  $r_{int} > 6\% \rightarrow 9\%$

Price of money is too high  $\rightarrow$  excess cash

$\rightarrow$  not enough bonds  $\rightarrow$  Bond prices  $\uparrow$ , Int. Rate  $\downarrow$

Imagine  $r_{low} < 6\% \rightarrow 3\%$

Price of money is too low  $\rightarrow$  not enough cash

$\rightarrow$  Too many bonds  $\rightarrow$  Bond price  $\downarrow \rightarrow$  int. rate  $\uparrow$

# Relationship between bond prices and interest rates

\$100, one-year bond

Bond Value = \$100, Bond Price < \$100

Bond Price = \$80 today

Return =  $100 - 80 = \text{Value} - \text{Price} = \$20$

$$\begin{aligned}\text{Interest Rate} &= \frac{\text{Return on Bond}}{\text{Price}} \times 100 \\ &= \frac{20}{80} \times 100 = 25\%\end{aligned}$$

Great Return, want more bonds  $\Rightarrow$  Excess Demand for Bonds!

$$\text{New Price} = \$90 \Rightarrow \text{Int. Rate} = \frac{100 - 90}{90} \times 100 = 11.1\%$$

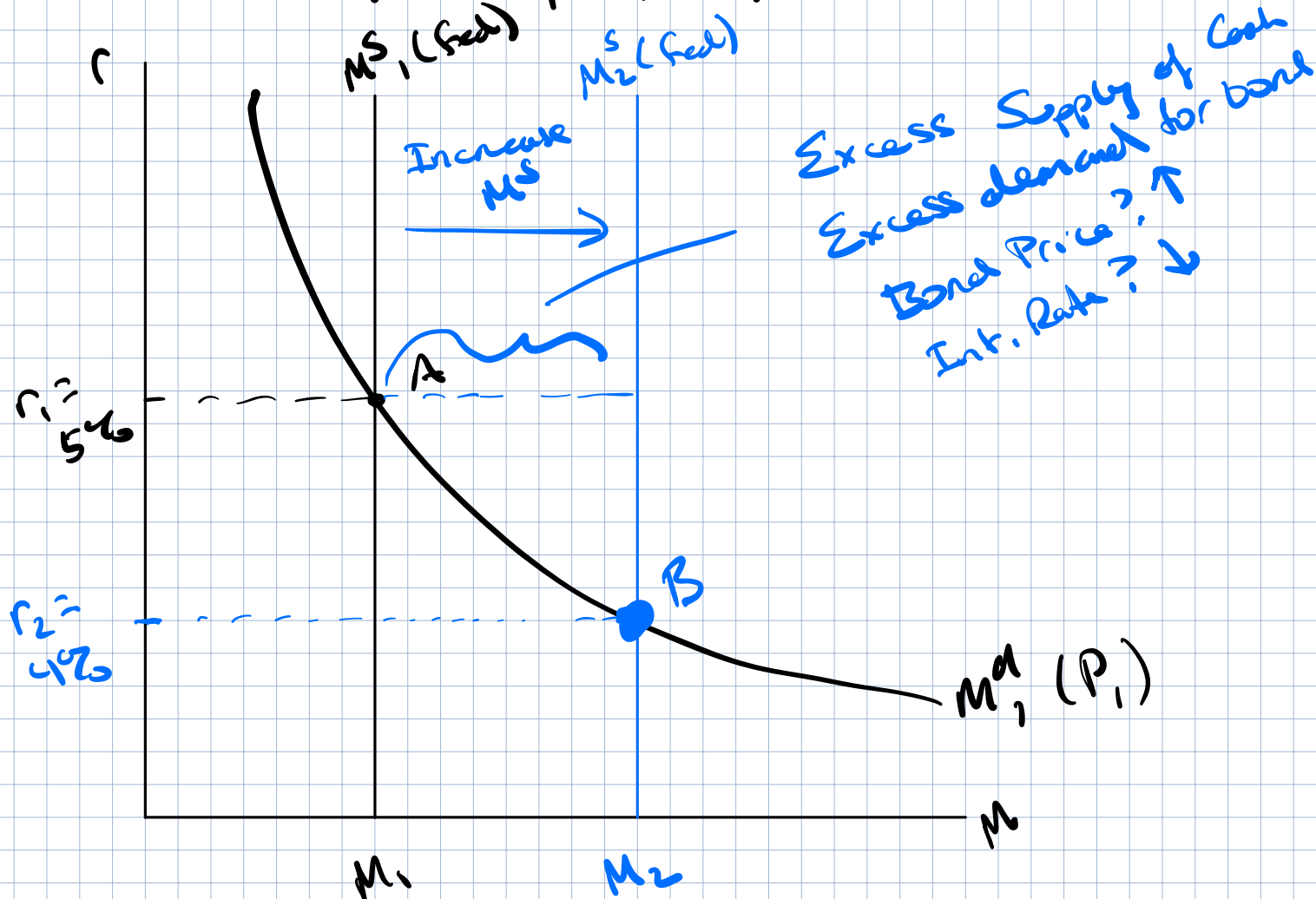
Bond Price  $\uparrow$ , interest rate  $\downarrow$   
Bond Price  $\downarrow$ , int. rate  $\uparrow$

Bond Price = \$475, Bond Value = \$500

$$\text{Int. Rate} = \frac{500 - 475}{475} \times 100 = 5.3\%$$

# Connect Money Market to Monetary Policy

Start at  $r_1 = 5\%$ ,  $M^S = M_1$



Fed wants  $r_1 = 5\%$  to  $r_2 = 4\% \rightarrow$  action?

Fed cannot move  $M^d$  curve

$\rightarrow$  Only change  $M^S \rightarrow$  increase  $M^S$  from  $M_1$  to  $M_2$

Buy bonds  $\rightarrow$  excess supply of cash  $\rightarrow$  Bond Price  $\uparrow$

$\downarrow$   
Int. Rate  $\downarrow$

Exercise: Start at  $r_1 = 5\%$ , want  $r_3 = 6\%$ ?

Step-by-Step

Why does changing the interest rate matter?

Connect to AE model

$$AE = C + I^P + G + NX$$

If  $r_1 = 5\% \rightarrow r_2 = 4\% \rightarrow$  What parts of AE change?

1. Firms that borrow in order to invest  $\rightarrow I^P$ 
  - $\rightarrow$  Big machines
  - $\rightarrow$  New operating facility
  - $\rightarrow$  Lower interest rates  $\rightarrow$  cheaper to borrow

} interest paid on expensive capital

If  $r \downarrow, I^P \uparrow$ , If  $r \uparrow, I^P \downarrow$

2. Consumers buy with a loan
  - $\rightarrow$  Big Ticket items: Car, furniture, appliances, etc.
    - $\hookrightarrow$  Air Conditioner
  - $\rightarrow$  Durable Goods
  - $\rightarrow$  Autonomous Consumption, if  $r \downarrow, AC \uparrow$   
 $r \uparrow, AC \downarrow$

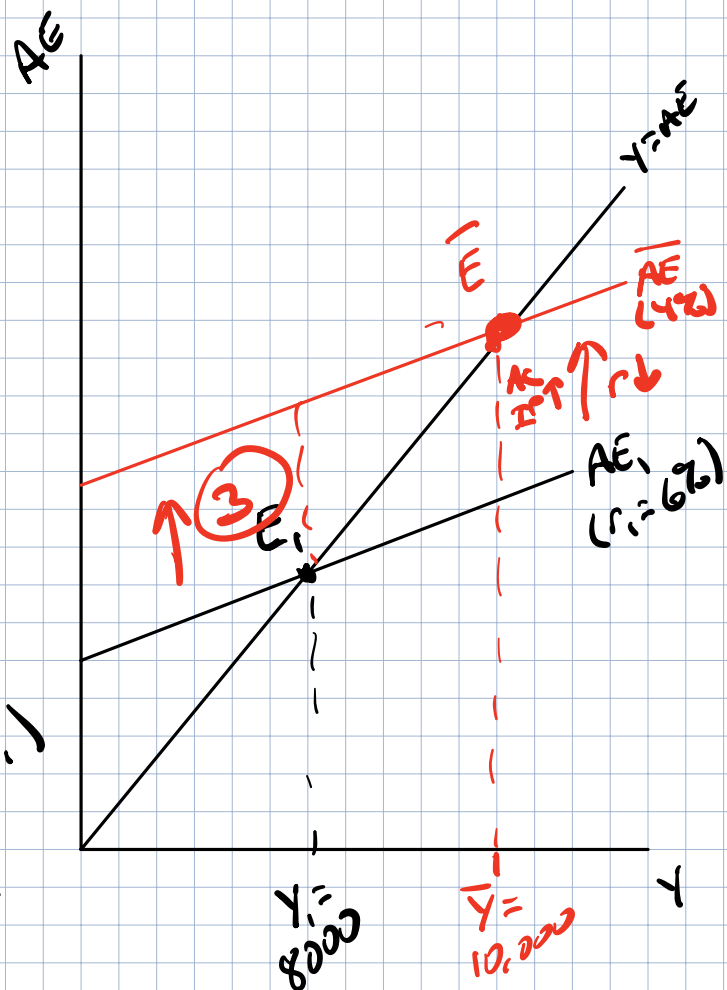
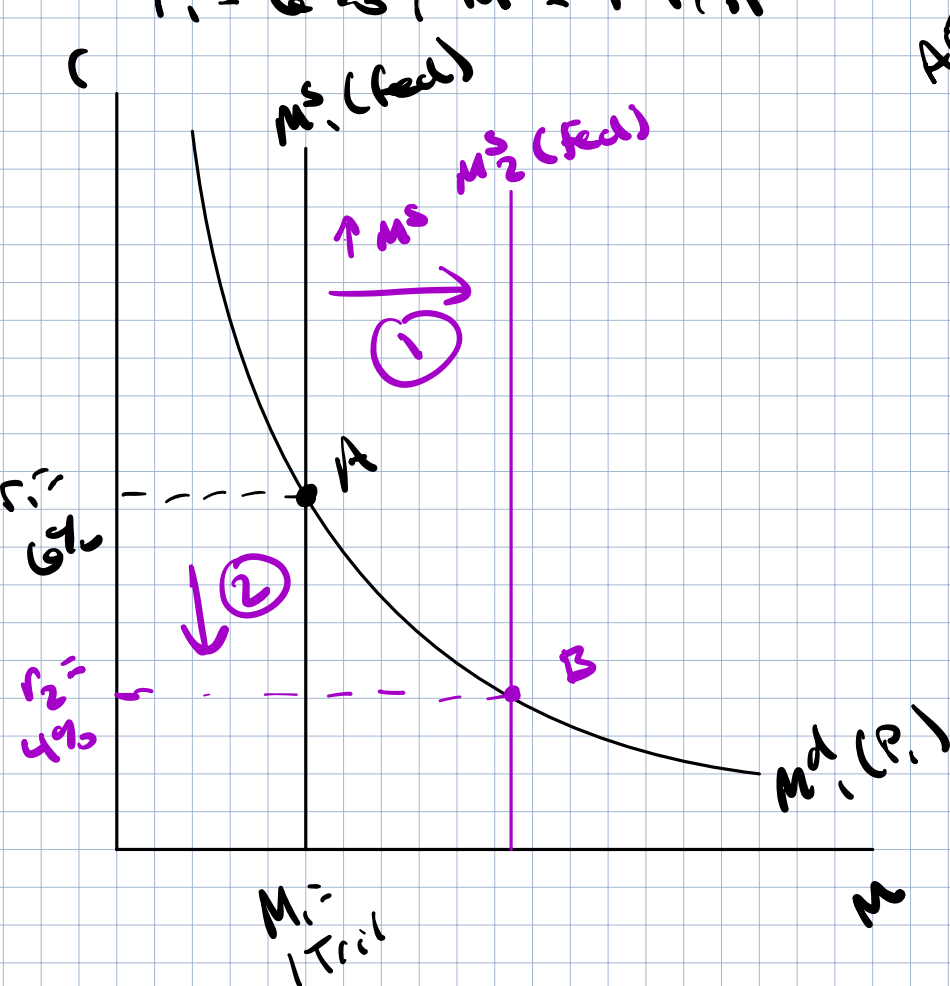
3. New Homes  $\rightarrow$  Developers to borrow to finance building  
As  $r \downarrow, I^P \uparrow$ ,  $r \uparrow, I^P \downarrow$

As  $r \downarrow \rightarrow AC \uparrow, I^P \uparrow \rightarrow AE \uparrow$

As  $r \uparrow \rightarrow AC \downarrow, I^P \downarrow \rightarrow AE \downarrow$

Current level of  $Y = 8,000$ ,  $\bar{Y} = 10,000$

$r_1 = 6\%$ ,  $M^S = 1 \text{ Tril}$



1. Fed buy bonds  $\rightarrow \uparrow M^S$
2. Decrease interest rate  $\Rightarrow r \downarrow$
3.  $\uparrow$  in  $AC, I^P \rightarrow \uparrow$  in  $AE$
4.  $\uparrow$  in  $Y \rightarrow Y_1$  to  $\bar{Y} = 10,000$ !